PROG7312

Task 1 Research

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Contents

User Engagement Strategies Considered 2
1. Real-time visual telemetry feedback 2
2. Proactive alert gamification 2
3. Simplified onboarding flows 2
4. Achievement based gamification 2
5. Prioritized Alert System 2

User Engagement Strategies Considered

1. Real-time visual telemetry feedback

Works by taking live sensor streams and compressing them into a visual indicator that is continuously recalculated so that operators can read the current system state with a single glance instead of parsing raw data [1] , and this allows operators to act on a mistake when it is correctable with the help of the system state feedback.

2. Proactive alert gamification

By adding gamification elements like daily streaks, virtual pets, or in-app unlocks to the operator dashboard, it gives fault-free operation a positive and visible signal, instead of just leaving it basic and silent [2]. This rewards the fault-free state and keeps operators engaged through long stretches of time where nothing is happening, and because the signal is updated on a predictable schedule it creates a habit out of a manual check-in [3].

3. Simplified onboarding flows

Configurations and diagnostic complexity are revealed in stages instead of all at once to make it digestible. New operators are only shown the parts that are relevant to their current work before introducing advanced options, which keeps the amount of new information that operators need to process at once to a manageable level [4]

4. Achievement based gamification

By introducing earnable points or completion feedback to an engineering task, like sensor registration or configuration validation, gives operators motivation and a feeling of closure to a task that would otherwise be seen as routine and mundane [5].

5. Prioritized Alert System

By structuring the dashboard in a way that filters, prioritizes and highlights only actionable alerts, it conditions operators into seeing alerts as important and urgent, which addresses the issue of operators being desensitized by high amounts of unimportant notification, which is a problem that is widely recognized in security and IoT operations [6].

Chosen Strategy: Telemetry Pulse

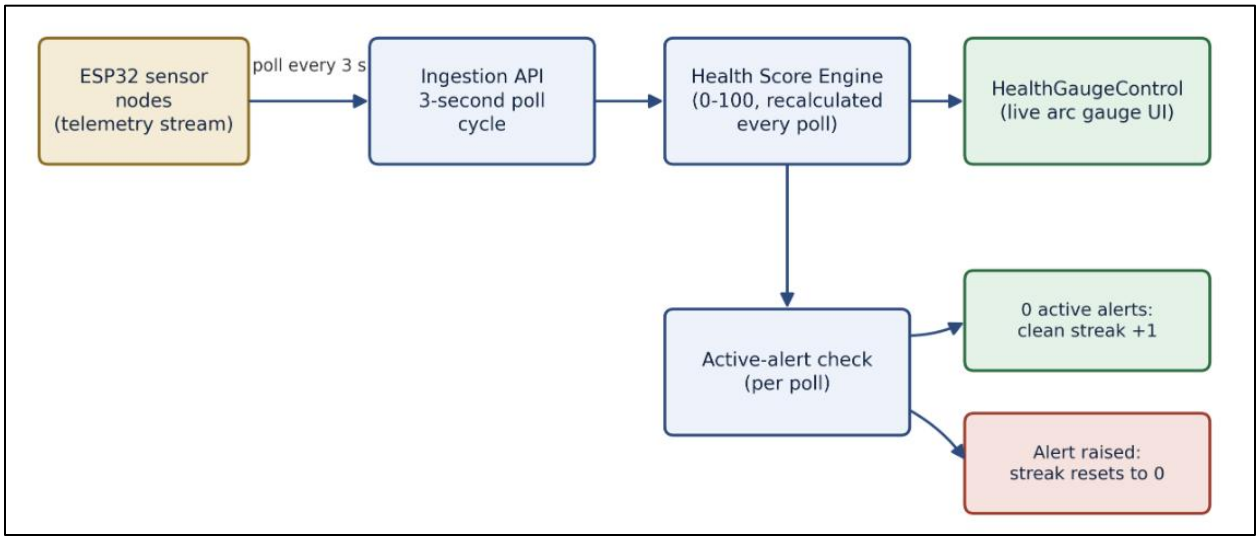
After researching the 5 strategies and considering which would be implemented, it was decided that Smart-X would implement a hybrid of real-time telemetry feedback and proactive alert gamification, done under a single widget named Telemetry Pulse. The widget has two halves that both share one data source, which is the 3 second telemetry poll. A arc gauge, called HealthGaugeControl, displays health from a range of 0-100 that gets recalculated on every poll cycle directly from live sensor reading. Next to it sits a clean-streak counter, which increases once per poll while zero alerts are active and resets to zero once a fault appears.

The two parts address the same problem but in different ways. The gauge communicates the magnitude, showing how far the system currently is from a healthy streak simply by looking without having to read raw telemetry values. This matches what [1] make for real-time IoT dashboards, that raw sensors stream is basically unusable without a compressed representation of the current state that is easy to read, and the representation has to update as fast as the data does to stay trustworthy. Because the gauge is tied to the same 3-second poll that ingests telemetry, a spike or disconnect that collapse the score would be visible in one cycle, which minimizes the time between something going wrong and the operator noticing. This is important because feedback loses a lot of value if it arrives after the fact rather than while the problem is still correctable [7].

The streak addresses the problem of alert fatigue. In [6], it identifies dashboard design as one way of reducing the fatigue and desensitization that sets in when operators are either seeing nothing or too much. A standard console gives no visual weight to there being nothing wrong in the system, so operators stop paying attention, and this causes them to miss a problem when it occurs. Because the streak increases on the same fixed timer, the repeated exposure turns the process of checking the health into a automatic process for the operator [3].

This also mirrors the effect found by [2], whereby adding game-like elements to an industrial control room dashboard, operators rated the gamified version as more engaging with little to no loss during monitoring, which shows that the design isn't just purely decorative [5].

When used together, the gauge and the streak give an operator two independent readings on the same cadence, with how bad it is right now, and how long it has been fine. A single blip that corrects itself will look different than a genuine negative trend, because the gauge will dip but the streak keeps climbing. When a problem is sustained, however, it will show both a low gauge value and a broken streak, which lets the operator easily distinguish when a real anomaly occurs without opening the raw log.



References

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